

COMPOSE: GENERATOR MATCHING OVER STOCHASTIC REWRITES FOR TRANS-DIMENSIONAL MOLECULAR GENERATION AND CONTROL

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ABSTRACT

Most generative models pick a generic corruption process (noise, an absorbing mask, a categorical flip) and learn to reverse it. The corruption is chosen for analytical convenience, its individual steps carry no executable meaning for the object being generated, and that object exists only at the end of the trajectory. We introduce *Rewrite Generator Matching* (RGM), which performs Generator Matching over stochastic rewrites and so defines the generative process as a stochastic program of executable rewrites: every state carries a state-dependent set of applicable marks, a learned hazard says how fast an event fires, a learned mark law says which one fires, and Generator Matching fits both from endpoint-conditioned rewrite programs. We instantiate RGM as COMPOSE, a *trans-dimensional* molecular process on $\mathcal{X} = \bigsqcup_{n=0}^{N_{\max}} \mathcal{X}_n$ with a distinguished null source at $n = 0$. Its marks create and destroy atoms, relabel and revalence them, reroute subgraphs, and close and open rings, so that molecular size, composition, connectivity, and topology are modelled jointly and every non-null committed state is a complete molecule. Execution and canonicalization aggregate equivalent marks (symmetry aliases, alternative addresses, distinct executable lowerings) into a canonical molecular-successor kernel. Under successor-preserving re-encodings that preserve aggregate successor rate, both the induced state process and fiber-constant successor-level controllers are invariant. Control then acts on that kernel rather than being attached afterwards: hard requirements delete transitions, and the finite-horizon Doob transform reweights the ones that remain, realizing exact terminal tilts and exact continuation after a mid-trajectory objective change wherever the state space is enumerable. Graph validity is not synthetic feasibility, and we claim no exactness at scale.

1 INTRODUCTION

Modern generative modeling rests on a standard move: pick a generic corruption process (Gaussian noise, an absorbing mask, a uniform categorical flip), apply it to an ambient encoding of the data, and learn to reverse it. The corruption is chosen for analytical convenience, its steps mean nothing about the object being generated, and that object typically exists only at the end of the trajectory. We ask instead: *can the generative process itself be a stochastic program of executable rewrites, with learned rates over the actions that are meaningful for the domain?* Where a domain already has a vocabulary of legal edits and an executor, that vocabulary can be the infinitesimal structure of a learned generative process.

We formulate this as *Rewrite Generator Matching* (RGM): Generator Matching over stochastic rewrites, carried out on an executable rewrite system. Fix typed rewrite rules and an executor. Every state x carries a state-dependent set $\mathcal{A}(x)$ of *applicable marks* (a rule, a concrete match, a typed payload), guarded by application conditions so that the executed successor is again a legal state. RGM learns a *total hazard* $\lambda_\theta(x, t)$, how fast an event fires, and a *conditional mark law* $p_\theta(a \mid x, t)$, which one fires; their product is a marked jump rate. The trained object is therefore a stochastic process over executable marks. Because several marks can execute to the same object, execution and canonicalization push the marked law forward to a kernel over distinct successor *objects*; that induced state-level process is the molecular control substrate. Generator Matching (Holderrieth et al., 2025) supplies the fitting principle: endpoint-conditioned rewrite programs give conditional

generators whose conditional expectation given the current state is the marginal generator, recovered by Poisson–Bregman regression on rates. Inference needs no endpoint, program, candidate bank, beam, or terminal repair.

Molecules are where this pays off most sharply, and COMPOSE is our instantiation. Its state space is *trans-dimensional*: for $n \geq 1$, $\mathcal{X}_n$ contains connected, valence-valid molecular graphs with n active heavy atoms, while $\mathcal{X}_0 = \{\emptyset\}$ contains a distinguished reversible null source that is not a molecule. COMPOSE generates on $\mathcal{X} = \bigsqcup_{n=0}^{N_{\max}} \mathcal{X}_n$, and its marks are classified by their effect on the stratum: *birth* ($\mathcal{X}_n \rightarrow \mathcal{X}_{n+1}$), *death* ($\mathcal{X}_n \rightarrow \mathcal{X}_{n-1}$, with a connected non-null remainder or the registered one-atom-to-null transition), and *same-cardinality* (relabel and revalence an atom, change a bond order, reroute a subgraph, close or open a cycle, restate a ring’s electronics). Size, composition, connectivity, and topology are therefore modelled jointly, and a trajectory can overshoot and correct rather than committing to a cardinality before generation begins. Because the rewrites are guarded, *every non-null step is a molecule and every transition is an executable rewrite*: at each non-null committed state, an oracle query, a hard constraint, or a chemist’s inspection is meaningful. Everything that follows (pathwise constraints, mid-trajectory intervention, branching, exact retargeting) uses that property, and we do not restate it.

The causal chain. One thesis runs through the paper, and each arrow is a section: *executable state-dependent rewrites* $\rightarrow$ *Generator Matching* (Section 3) $\rightarrow$ *a learned trans-dimensional molecular process* (Section 4) $\rightarrow$ *a canonical molecular-successor kernel* (Section 5) $\rightarrow$ *exact and dynamic stochastic control* (Section 6). The order matters: most of what looks like a chemistry capability is a property of the kernel, and the kernel is a property of the rewrite system. Source-conditioned lead optimization is the flagship *application* (experiment G), not the definition of the framework.

What we do not claim as new. Atom insertion and deletion, valid molecular editing, Generator Matching, Doob transforms, variable-dimension generation, and graph grammars are all established; Section 2 places each. The defensible claim is their formal combination and interface: a generator-matched stochastic graph-rewrite process whose state-dependent marks create, delete, relabel, reconnect, and cyclize molecular structure; whose legal support is closed over molecular states; whose equivalent action encodings are quotiented into a molecular kernel; and whose resulting process supports exact finite-horizon control.

1. **Generator Matching over stochastic rewrites:** Generator Matching over state-dependent executable rewrite marks, with a learned total hazard and conditional mark law fitted by a Poisson–Bregman objective from certified endpoint-conditioned programs. Stated for any executable rewrite system, not only for chemistry.
2. **Trans-dimensional molecular generation and the canonical successor quotient:** a process on $\bigsqcup_{n=0}^{N_{\max}} \mathcal{X}_n$ that creates and destroys typed graph entities and rebuilds ring topology compositionally, with every non-null committed state molecular; equivalent marks are aggregated into a molecular kernel, which we prove is invariant under successor-preserving re-encodings that preserve aggregate successor rate, as are fiber-constant successor-level controllers.
3. **Control of the resulting molecular kernel:** the shared framework supports separately trained *de novo* and source-conditioned editing processes; on an induced editing kernel we give exact finite-horizon Doob control with exact continuation after a mid-trajectory objective change and pathwise support constraints, and we specify a goal-conditioned learned controller that will be evaluated and calibrated as an approximation.

2 RELATED WORK AND POSITIONING

Generator Matching and graph generative processes. Generator Matching (Holderrieth et al., 2025) regresses a marginal generator from tractable endpoint-conditional generators; it prescribes no state space, transitions, or coupling, and RGM supplies those for rewrite systems. Discrete graph diffusion and flow models corrupt node and edge categories and learn a denoiser (Vignac et al., 2023; Austin et al., 2021; Campbell et al., 2022; Qin et al., 2025); their intermediates are discrete graphs but need not be connected or valence-admissible, so an oracle query at an intermediate is not chemically defined.

[Placeholder: schematic to be drawn; panel specification in the caption.]

$$\begin{array}{ll}
 \text{(a)} \quad x \mapsto \mathcal{A}(x) \xrightarrow{\lambda_\theta(x,t), p_\theta(a|x,t)} q_\theta(a|x,t) & \text{(b)} \quad \mathcal{X}_{n-1} \xleftarrow{\text{death}} \mathcal{X}_n \xrightarrow{\text{birth}} \mathcal{X}_{n+1}, \\
 & \mathcal{X}_0 = \{\emptyset\} \\
 \text{(d)} \quad P_\theta^{\text{raw}}(y|x,t) = \sum_{a \in \mathcal{G}_y(x)} p_\theta(a|x,t), \quad P_\theta \propto \mathbf{1}\{y \in \mathcal{S}^+(x)\} P_\theta^{\text{raw}} & \text{(f)} \quad \text{timed} \\
 & \text{CTMC} \mid \text{fixed-budget jump chain}
 \end{array}$$

Figure 1: **Generator Matching over stochastic rewrites, and the trans-dimensional COMPOSE process.** The abstraction (applicable set, learned hazard, conditional mark law); the molecular instantiation as strata $\mathcal{X}_n$ with birth arrows up, death arrows down, and same-cardinality edges within a stratum; canonical successor aggregation, drawn as several distinct raw marks converging on one molecular successor; compositional ring support; the two inference regimes; and the control interface. Panels specified in Appendix A.

Variable cardinality and trans-dimensional processes. Trans-dimensional generative modeling via jump diffusion learns processes moving between spaces of different dimension (Campbell et al., 2023), the closest formal precedent for our state space, and the reason we claim no novelty for variable-dimension generation; the distinction is that our dimension changes are *executable typed graph edits* with application conditions rather than jumps in an ambient space. GrIDDD adds monotone node insertion and deletion (Ninniri et al., 2025), Edit Flows builds insertion/deletion/substitution flows (Havasi et al., 2025), Morph changes 3D cardinality by unbalanced optimal transport (Franke et al., 2026), and DDSBM learns CTMC bridges for graph transformation (Kim et al., 2024). COMPOSE differs in that one jump is a guarded chemical rewrite between complete molecules, in both directions of cardinality, with topology-changing operators alongside.

Validity by construction, editors, and control. Constrained graph diffusion projects onto restricted property classes: edge-deletion-invariant topology for ConStruct, aggregate counts for PRODIGY, fixed formula and degree for CoCoGraph (Madeira et al., 2024; Sharma et al., 2024; Ruiz-Botella et al., 2026); molecular grammars generate through legal productions (Kajino, 2019; Guo et al., 2022); reaction-based GFlowNets traverse executable transformations (Kozarski et al., 2024; Zhu et al., 2026). Stochastic graph rewriting is the closest operational precedent, defining matches, application conditions, and a CTMC from contextual propensities (Danos et al., 2015; Behr et al., 2021); we import that rule/rate separation and replace fixed propensities with a learned time-inhomogeneous generator. A separate line edits valid molecules directly (Xie et al., 2021; Fu et al., 2021; Jensen, 2019; Gao et al., 2022); we claim no novelty there. MARS and MIMOSA couple learned or structured edit proposals to task-conditioned MCMC-style optimization, while COMPOSE learns an objective-independent reference transport that is subsequently controlled. Doob h -transforms are classical, and inference-time controllers are instances of Ψ in (10), which we run on our own kernel so a comparison isolates the control rule rather than the base model. Appendix C expands each of these and Table 3 is the property-by-property ledger.

3 GENERATOR MATCHING OVER STOCHASTIC REWRITES

RGM is stated for any executable rewrite system; we develop it on the molecular state space that Section 4 instantiates. Generic objects: the state set, applicable marks, hazard, mark law, successor aggregation, and objective. Chemistry-specific: the grammar, rules, executor, and source processes.

States, marks, and the applicable set. Given a declared grammar Γ and an entity cap $N_{\max}$, let $\mathcal{M}_\Gamma = \{\emptyset\} \sqcup \{G : 1 \leq |V(G)| \leq N_{\max}, G \text{ connected and admissible under } \Gamma\}$. The distinguished null state $\emptyset$ is a reversible source, not a molecule. A rule $r \in \mathcal{R}$ has a left pattern, a right pattern, and application conditions; a *mark* $a = (r, m, \xi)$ is a rule, a match $m \in \text{Match}_r(G)$, and a typed payload, and its executor T_a is partial. The *applicable set* is

$$\mathcal{A}(G) = \{(r, m, \xi) : m \in \text{Match}_r(G), \text{Legal}(G, r, m, \xi) = 1\}, \quad T_a G \in \mathcal{M}_\Gamma \quad \forall a \in \mathcal{A}(G), \quad (1)$$

so legality is state-dependent and closure holds by construction rather than by post-hoc repair.

The two learned objects. RGM factorizes the marked rate into a scalar hazard and a conditional mark law,

$$q_\theta(a \mid G, t) = \lambda_\theta(G, t) p_\theta(a \mid G, t), \quad \lambda_\theta(G, t) \geq 0, \quad (2)$$

$$\sum_{a \in \mathcal{A}(G)} p_\theta(a \mid G, t) = 1 \quad \text{when } \mathcal{A}(G) \neq \emptyset$$

We adopt the terminal convention $\lambda_\theta(G, t) = 0$, $q_\theta(\cdot \mid G, t) = 0$ when $\mathcal{A}(G) = \emptyset$; the conditional law is then unused. The hazard is a rate in events per unit operational time and governs *when*; the mark law governs *which*. Only a conditioned mark law is needed to walk the productive embedded jump chain. The editing and *de novo* regimes therefore share this factorization and executor, but use separately fitted checkpoints and different inference semantics (Section 4.3). The marked generator is $(L_{\theta,t}f)(G) = \sum_{a \in \mathcal{A}(G)} q_\theta(a \mid G, t)[f(\tau_a G) - f(G)]$. Since several marks may execute to the same object, we write $\mathcal{G}_H(G) = \{a \in \mathcal{A}(G) : \tau_a G \simeq H\}$ for the *successor fiber* of H and aggregate,

$$\mathcal{S}(G) = \{H : \mathcal{G}_H(G) \neq \emptyset\}, \quad Q_{\theta,t}(G, H) = \sum_{a \in \mathcal{G}_H(G)} q_\theta(a \mid G, t), \quad (3)$$

$$P_\theta^{\text{raw}}(H \mid G, t) = \sum_{a \in \mathcal{G}_H(G)} p_\theta(a \mid G, t). \quad (4)$$

The raw pushforward includes every executable mark, including canonical self or implementation-virtual events. Let $\mathcal{S}^+(G) \subseteq \mathcal{S}(G)$ be the distinct successors admitted by the registered productive-commit predicate, which excludes those events, and define

$$Z_\theta^+(G, t) = \sum_{J \in \mathcal{S}^+(G)} P_\theta^{\text{raw}}(J \mid G, t), \quad (5)$$

$$P_\theta(H \mid G, t) = \frac{\mathbf{1}\{H \in \mathcal{S}^+(G)\} P_\theta^{\text{raw}}(H \mid G, t)}{Z_\theta^+(G, t)} \quad \text{when } Z_\theta^+(G, t) > 0.$$

Thus P_θ^{raw} is the executor-and-canonicalization pushforward of the mark law, while P_θ is the productive embedded molecular jump kernel used for editing. If $Z_\theta^+ = 0$, the production sampler halts and records a stall; finite-horizon algebra represents that declared terminal convention with an absorbing row rather than renormalizing nonexistent productive mass. Section 5 develops both objects as first-class constructions.

Latent rewrite programs as conditional paths. Sample a source–target pair from a coupling with marginals π_0 and p_{data} , and let a randomized compiler emit a certified program $Z = (G_0, a_1, \dots, a_K, G_K^Z)$ with every $G_k^Z \in \mathcal{M}_\Gamma$ and $G_K^Z \simeq G_1$. Conditional on Z , define a progress variable K_t with a binomial marginal and $X_t = G_{K_t}^Z$; the pure-birth realization fires only the next instruction, giving a conditional marked rate $r_t^Z(a \mid k)$ that is nonzero only for $a = a_{k+1}$ and, in operational time, equals the number of remaining instructions (Appendix H). Sampling k from its marginal avoids simulating the teacher chain. Writing $R_t^Z(H \mid X_t, k) = \sum_{a \in \mathcal{G}_H(X_t)} r_t^Z(a \mid k)$, the object to be learned is the marginal generator $Q_t(G, H) = \mathbb{E}[R_t^Z(H \mid X_t, K_t) \mid X_t = G]$, a conditional expectation inherited from Generator Matching; the rewrite state space, program construction, applicable sets, and successor aggregation are what RGM contributes.

Objective. For $\phi(u) = u \log u - u$, the Bregman divergence between nonnegative rates gives, up to target-only constants,

$$\mathcal{L}_{\text{RGM}}(\theta) = \mathbb{E}_{t,Z,K_t} \left[\sum_{H \in \mathcal{S}(X_t)} \left\{ Q_{\theta,t}(X_t, H) - R_t^Z(H \mid X_t, K_t) \log Q_{\theta,t}(X_t, H) \right\} \right]. \quad (6)$$

The first term calibrates total firing mass, the second rewards the teacher successor, and unlike a classification loss over the observed action it is proper for rates.

Three objective identities must remain distinct. Equation (6) is the general successor-rate population objective used for the RGM consistency result below. The completed 16,000-step diagnostic run instead optimized the implemented mark-level objective

$$\mathcal{L}_{\text{hist}} = \mathbb{E} \left[\lambda_\theta(X_t, t) - r_t^Z \{ \log \lambda_\theta(X_t, t) + \log p_\theta(a^* \mid X_t, t) \} \right], \quad (7)$$

with the teacher mark a^* selected by the compiled program (and graft aliases already grouped). The editing-V2 lane declares productive canonical-successor identity training,

$$\mathcal{L}_{\text{edit-V2}} = \mathbb{E}[-w \log P_\theta(H^* \mid X_t, t)] + \alpha_{\text{haz}} \mathcal{L}_{\text{haz}}, \quad (8)$$

where H^* is the teacher molecule, w is the frozen sampling correction, and any hazard loss and coefficient are separately registered and checkpoint-bound. The editing identity term does not silently inherit the historical mark objective, and the timed *de novo* lane retains its own calibrated rate objective.

Assumption 1 (Finite closed rewrite system, consistent programs). $N_{\max}$ and all type vocabularies are finite; every committed action satisfies (1); the total rate is bounded on the finite horizon; and the program sampler has endpoint marginals $G_0 \sim \pi_0$, $G_K^Z \sim p_{\text{data}}$ with every sampled program executable and terminating at its target.

Proposition 2 (Population minimizer; endpoint law). *Under assumption 1, for every (G, t) occupied by the program mixture and every $H \in \mathcal{S}(G)$, the unrestricted nonnegative-rate minimizer of (6) is the marginal aggregated rate $Q_t(G, H)$ defined above. If in addition $Q_{\theta, t} = Q_t$ on occupied states, and the learned process starts at π_0 , then its time marginals equal the program-mixture marginals and its terminal law is p_{data} .*

In words: fitting rates by (6) recovers exactly the state-level generator of the program mixture, and an exactly-fitted model transports π_0 to p_{data} . Neither statement is chemistry-specific, and neither implies that a finite network attains the population minimizer or that teacher programs are optimal paths. *Sketch:* the objective decouples across (G, t, H) into $u - \mathbb{E}[Y] \log u$, strictly convex with minimizer $\mathbb{E}[Y]$; the endpoint claim follows from uniqueness of the finite-state forward solution (Appendix G).

The framework is now fixed; what remains is to choose the rewrite system, and that choice is where the molecular content lives.

4 COMPOSE: A TRANS-DIMENSIONAL MOLECULAR REWRITE PROCESS

4.1 A TRANS-DIMENSIONAL STATE SPACE

Molecules of different sizes are different objects, and a process honest about that must move between spaces of different dimension. For $n \geq 1$, let $\mathcal{X}_n$ be the connected, valence-admissible molecular graphs with exactly n active heavy atoms; let $\mathcal{X}_0 = \{\emptyset\}$ contain the distinguished null state. Then

$$\mathcal{X} = \bigsqcup_{n=0}^{N_{\max}} \mathcal{X}_n, \quad (9)$$

so that the state space $\mathcal{M}_\Gamma$ of Section 3 is exactly $\mathcal{X}$. The null graph is not a molecule, but production treats it as a reversible source, $\emptyset \rightleftharpoons$ one-atom molecules. Every applicable mark is a *birth* ($\mathcal{X}_n \rightarrow \mathcal{X}_{n+1}$), a *death* ($\mathcal{X}_n \rightarrow \mathcal{X}_{n-1}$), or *same-cardinality* ($\mathcal{X}_n \rightarrow \mathcal{X}_n$), and Table 1 classifies the operator registry accordingly.

The slot array is a coordinate system, not the state. The implementation stores a state in a fixed array of $N_{\max}$ persistent slots, and it matters that the array is a *coordinate system* while the semantic state is the *active molecular graph*. A birth does not flip a padding indicator: it creates a typed graph entity that participates in valence, connectivity, ring perception, and canonicalization, and the successor is a different molecule in a different stratum. Two states with the same active subgraph in different slots are the same state, and the canonical successor map of Section 5 performs that identification. Persistent slots are semantic action *addresses*: serialized trajectories therefore store the exact slot-indexed graph state rather than reconstructing it from a canonical string, which preserves molecular identity but not action coordinates. The generative consequence is that cardinality is not a prior draw: fixed-cardinality processes commit to n before generating, insertion-only processes can grow but never revise, whereas this process adapts cardinality to what the trajectory has built, including to an objective supplied only at inference. Experiment C (Table 2) tests exactly that.

4.2 OPERATORS, SCOPE, AND RINGS

The bounded rebuild currently tests eight development families: atom insertion, deletion, and restatement; bond reordering; graft; cycle closing and opening; and coordinated ring-system restatement. Whole-ring deletion and the legacy whole-ring growth macro are disabled in this pilot. This *Active8*

Table 1: The rewrite classes, by their action on the strata of (9). The complete typed registry, applicability conditions, and inverse relations are given in Appendix D.

Rewrite class	Cardinality	Examples
Birth	$\mathcal{X}_n \rightarrow \mathcal{X}_{n+1}$	atom insertion
Death	$\mathcal{X}_n \rightarrow \mathcal{X}_{n-1}$	atom deletion
Structural	$\mathcal{X}_n \rightarrow \mathcal{X}_n$	atom restate, bond reorder, graft (reroute), cycle close/open, ring aromaticity restate

basis is not yet the final production support: the operator set is frozen only after registered reachability, successor-level capacity, and short-pilot evidence, and any macro must earn its place by a measured path-efficiency gain. Cardinality and topology are therefore tested separately rather than inferred from operator counts. *Graft* remains load-bearing because it relocates a bridge-attached fragment in one atomic transaction and supplies short compositional paths for attachment and linker changes without exposing a disconnected state. The declared scope ($N_{\max} = 40$ active atoms, 15 neutral (element, valence) classes over C/N/O/F/S/P/Cl/Br/I/B, charge carried rather than designed, no stereochemistry) is in Appendix D; it admits 800/800 of the benchmark leads where a neutral C/N/O/F vocabulary admits 33.0%.

Rings are constructed, not looked up. Ring topology is supported *compositionally*: `cycle_close` inserts one legal bond between two nonbonded atoms and raises graph cycle rank by one, while `cycle_open` deletes one legal non-bridge cycle edge and lowers it by one. At the graph-topology level, every connected graph is a spanning tree plus its non-tree edges, so repeated close/open operations do not define support through a finite ring-template catalog. This graph statement does not prove a valence-valid path to every chemically labelled ring system within a finite edit budget; those claims require constructive executor replay and the topology tasks of experiment D. The legacy whole-ring growth macro remains disabled. Ring-system restatement is retained provisionally for coordinated electronic changes, subject to the same reachability and capacity gates; whole-ring deletion is disabled in the bounded pilot. A future ring macro may be tested as an explicit efficiency proposal, never as the definition of ring support.

4.3 SOURCE PROCESSES AND TWO SAMPLING REGIMES

The network is source-agnostic (it reads only the current state and model time), so a source molecule is an initial condition, not a repeated network input. Editing and *de novo* generation are nevertheless separate fitted regimes. The editing prior is trained from real-endpoint analogue programs and executor-generated support paths, begins from a supplied molecule, and is sampled as a **fixed-budget embedded molecular jump chain**: each step draws from the productive canonical-successor law, while continuous-time hazard magnitude is discarded. The unconditional lane requires a separate broad-organic checkpoint, a frozen *de novo* source distribution including the reversible null source, nonzero *de novo* training mass, and calibrated hazard. Its inference is a **timed CTMC**: evaluate λ_θ , draw an exponential waiting time, sample a masked mark, commit, and recompute. The shared framework and executor do not make the editing checkpoint evidence for unconditional generation. For editing, the exact control object is indexed by the *remaining budget* rather than by time (Section 6).

Proposition 3 (Closure and trans-dimensional reachability). *Under assumption 1, if $G_0 \in \mathcal{M}_\Gamma$ then every state committed by the teacher or the learned process lies in $\mathcal{M}_\Gamma$ almost surely. Moreover, ignoring chemical labels, any two finite connected simple graphs are joined by a finite sequence of leaf births and deaths and non-bridge edge insertions and deletions with every intermediate connected; $\emptyset$ is joined reversibly to one-vertex graphs; and any cyclic topology is reached from an acyclic precursor by inserting its non-tree edges one at a time.*

In words: committed states cannot leave the declared support, and the abstract graph basis is trans-dimensional and topology-changing. The null state is the sole non-molecular exception. The reachability half is a graph statement only: in COMPOSE every non-null step must additionally satisfy the molecular grammar, so a valence-feasible route between two *particular* molecules is not implied, and it says nothing about finite-budget path cost. Dataset-pair reachability is therefore certified

constructively by compiler replay (experiment B) rather than inferred from Proposition 3. Proof in Appendix G.

Both halves of Proposition 3 are statements about *states*. Making them true of the learned process, rather than of an arbitrary encoding of it, requires one more construction.

5 THE CANONICAL MOLECULAR-SUCCESSOR QUOTIENT

Equations (4) and (5) are definitions, but they are better read as decisions about the state-level object induced by the marked process. The successor fibers partition $\mathcal{A}(G)$, and four consequences follow. Aggregation removes symmetry and address multiplicity: deleting either of two equivalent methyls, or naming one Kekulé structure two ways, is one transition named twice, and without aggregation a molecule receives probability in proportion to how many ways the encoding can name it. It defines the pushed-forward process on molecular states rather than on the slot array of Section 4.1; under the conditions below it makes controllers encoding-independent, since a controller sees only one scalar per successor; and it lets primitive and macro routes to one successor contribute to a single transition. The production editing kernel additionally conditions away canonical self and virtual events rather than counting them as fixed-budget edits. The completed diagnostic run optimized selected executable marks, except that graft already used a log-sum-exp over its canonical-successor aliases. Its molecular law is therefore the executor-and-canonicalization pushforward of the trained mark law; we do not retroactively describe that run as successor-trained. Equation (8) is the declared replacement objective, but becomes a completed method claim only after its packed implementation and scaling gates pass.

It is tempting to say that any refinement or coarsening of marks preserving aggregate successor rates leaves the process and its controlled law unchanged. That is true only under two conditions the informal statement hides, and false for a practically important class of controllers.

Definition 4 (Successor-preserving, rate-preserving re-encoding). Two mark systems $(\mathcal{A}, T, q)$ and $(\mathcal{A}', T', q')$ on the same state space are so related if for every G there is a surjection $\varphi_G : \mathcal{A}'(G) \rightarrow \mathcal{A}(G)$ with $T_{\varphi_G(a')}G \simeq T_{a'}G$ for all a' , so marks may be split or merged only *within* a successor fiber, never across fibers, and if $\sum_{a \in \mathcal{G}_H(G)} q(a | G, t) = \sum_{a' \in \mathcal{G}'_H(G)} q'(a' | G, t)$ for every G, H, t .

Proposition 5 (Encoding invariance, and its exact scope). *Let two mark systems be related by a successor-preserving, rate-preserving re-encoding, under assumption 1, and use the same state-level productive-commit predicate. Then (i) the state-level generators coincide, hence so do the total hazards $\lambda_\theta = \lambda'_\theta$, the molecular CTMC laws, the productive embedded jump kernels $P_\theta = P'_\theta$ wherever defined, all time marginals, and the terminal law; (ii) for any controller Ψ_t that is fiber-constant (a function of (G, H, t, ω) rather than of the mark), the controlled kernels of (10) coincide, hence so do the controlled molecular laws; (iii) hard support masks $\mathbf{1}\{C(H)\}$ and the finite-horizon Doob transform of Theorem 7 are fiber-constant by construction, so the values h_b and the controlled kernels are identical under any such re-encoding.*

In words: how the implementation names its edits is not part of the state-level model when the re-encoding preserves successors and aggregate rates. Two boundaries are load-bearing rather than pedantic. *Fiber-crossing merges are ill-posed*, not merely lossy: merging marks with non-isomorphic successors would stop the executor being a function of the merged mark, which Definition 4 excludes and the informal phrasing does not. And *mark-level controllers are not generally invariant*: if H is reachable by four symmetric aliases of rate $1/4$ under one encoding and by a single mark of rate 1 under another, a mark-level power tilt $\Psi \propto q(a)^{\beta-1}$ gives H controlled mass $4 \cdot (1/4)^\beta$ versus 1, differing for every $\beta \neq 1$. Top- k or nucleus truncation over marks and per-family tilts generally fail for the same reason; a per-family multiplier is invariant only if it is constant across every alias in each successor fiber. Our molecular controllers are written over successor states. The proposition concerns the pushed-forward process, not the training representation: mark-level supervision is generally not invariant to within-fiber refinement. Appendix G gives the proof and the full adversarial check; experiment E tests it empirically.

Because the modelled object is a kernel on molecules, a controller can now be defined without reference to the rewrite vocabulary at all.

6 FINITE-HORIZON STOCHASTIC CONTROL OF THE MOLECULAR KERNEL

Let C be a predicate on $\mathcal{M}_\Gamma$ and $\Psi_t(\cdot, \cdot; \omega) : \mathcal{M}_\Gamma \times \mathcal{M}_\Gamma \rightarrow [0, \infty)$ a controller with parameters ω . For the continuous-time construction, assume Ψ_t is locally bounded on the registered finite horizon over base-supported transitions. Controlled generation is

$$\tilde{Q}_t(G, H) = \mathbf{1}\{C(H)\} Q_{\theta,t}(G, H) \Psi_t(G, H; \omega), \quad H \neq G, \quad (10)$$

with the diagonal re-summed to the negative total outgoing rate. One expression covers hard requirements, preference tilting, Doob control ($\Psi = h(H)/h(G)$), source-conditioned editing ($G_0 = G_s$), mid-trajectory objective changes (time-varying ω_t), and unguided generation ($C \equiv 1, \Psi \equiv 1$), with controllers swapped without retraining.

Proposition 6 (Closure under control; pathwise invariance). *Under assumption 1, for any predicate C and any nonnegative controller Ψ_t satisfying the local-boundedness condition above, if $G_0 \in \mathcal{M}_\Gamma$ then every state committed by (10) lies in $\mathcal{M}_\Gamma$ almost surely, and the controlled support satisfies $\tilde{Q}_t(G, \cdot) \ll Q_{\theta,t}(G, \cdot)$. If moreover $C \equiv 1$ and $\Psi_t(G, H) > 0$ for every base-supported $H \neq G$, the off-diagonal supports coincide. Consequently, for a predicate P with $P(G_0)$ true, taking $C = P$ makes P an invariant: $P(G_k)$ holds at every committed G_k , not only at the horizon. If feasibility is evaluated before a separate downstream property oracle, that oracle need not be called on a state rejected by P .*

In words: a controller may delete or reweight transitions but can never create one, so guidance cannot leave the valid molecules, and a rule-closed constraint becomes a property of the path rather than a filter on the endpoint. This is what makes *pathwise* constraints expressible at all: the restriction $\mathcal{A}_P(G) = \{a : P(T_a G)\}$ can be formed only because $T_a G$ is a complete molecule on which P is decidable.

Finite-horizon Doob control. Editing walks a fixed- K -step embedded chain in which the hazard and clock are discarded (Section 4.3), so the exact control object is not the continuous-time generator transform: the desirability depends on the *remaining budget*.

Theorem 7 (Exact finite-horizon terminal tilt and exact continuation). *Let $\{P_{\theta,b}\}_{b=1}^K$ be row-stochastic transition kernels on a finite state space, indexed by remaining budget and including any explicitly declared absorbing terminal rows. A stationary embedded chain is the special case $P_{\theta,b} \equiv P_\theta$. Let $g \geq 0$ be a nonzero terminal desirability and define $h_0 = g$ and $h_b(x) = \sum_y P_{\theta,b}(x, y) h_{b-1}(y)$. At a state with $h_b(x) > 0$ and b steps remaining, let*

$$P_{\theta,b}^g(x, y) = P_{\theta,b}(x, y) h_{b-1}(y) / h_b(x). \quad (11)$$

Then $P_{\theta,b}^g$ is row stochastic and its support is contained in that of $P_{\theta,b}$. Let $\Pr$ denote the uncontrolled law generated by the budget-indexed sequence. From x_0 with budget K and $h_K(x_0) > 0$,

$$\Pr^g(X_K = y \mid x_0) = \Pr(X_K = y \mid x_0) \frac{g(y)}{h_K(x_0)}, \quad h_K(x_0) = \sum_y \Pr(X_K = y \mid x_0) g(y). \quad (12)$$

Replacing g by g' at an intermediate state x with b' steps remaining and $h_{b'}'(x) > 0$ yields exactly the g' -tilt of the continuation from (x, b') . A state with $h_b(x) = 0$ is unreachable under the conditioned law, and the transformed kernel is undefined there rather than numerically patched. Support equality holds at a controlled row only when $h_{b-1}(y) > 0$ for every base-supported successor y .

In words: with an exact desirability, a budget-conditioned reweighting of existing molecular transitions realizes the specified terminal distribution exactly, and changing the objective mid-trajectory is a fresh finite-horizon problem from the current molecule. The transform never creates a transition, but hard conditioning may prune base edges. Proof sketch: the h factors telescope along every reachable path, so all paths to a fixed endpoint are reweighted by the same $g(y)/h_K(x_0)$; Appendix G gives the argument and the continuation case.

Remark 8 (Exactly where exactness stops). Exactness is *budget-specific*: the controlled marginal after fewer than K steps is an h_{K-j} -tilt, not the terminal g -tilt, so early stopping does not inherit (12) and a budget-agnostic controller is not exact even with a perfect value. It requires an *exact* h and exact simulation; on the full molecular state space h is intractable, the controller is learned, and it will be reported as an approximation with a calibration curve and a backward-equation residual. And the terminal form covers endpoint desirabilities: genuinely path-dependent rewards need a Feynman–Kac potential, a killed generator, or an augmented state.

6.1 THE GOAL-CONDITIONED CONTROLLER

Exactness is available only where h can be computed. At scale we learn a budget-conditioned desirability

$$h_\phi(b, G; G_s, z, m), \quad (13)$$

with b the remaining edit budget, G_s the source lead, z a preference or target region, and m a protected-substructure mask, and set $\Psi_t = h_\phi(b - 1, H; \cdot) / h_\phi(b, G; \cdot)$ in (10). Three properties follow from earlier sections rather than from the controller’s design. It is *source-conditioned without a source-conditioned base*: G_s, z, m, b enter only through h_ϕ , so one base kernel serves every campaign. It is *representation-independent*: h_ϕ scores molecular successors, so it is fiber-constant and Proposition 5 applies. And it *retargets without restarting*: conditioning on b makes a mid-trajectory change of z the exact continuation problem of Theorem 7.

Three lanes stay distinct. The *exact* controller computes h by backward recursion, exists only on enumerable spaces, and is the ground truth of experiment F. The *learned* controller (13) is an approximation that will be evaluated and calibrated as one. *Reward fine-tuning*, changing θ so the base prefers high-reward regions, moves the reference process rather than reweighting it, forfeiting the guarantee that one base kernel serves all objectives; no result here depends on it.

7 EXPERIMENTS

Status of the empirical record. The first large-data scientific editing run completed 16,000 steps on 661,105 admitted training traces, with 32 snapshots and a complete run manifest. It is a scientifically valid diagnostic run, not a selected model. Its aggregate loss improved, but family-resolved diagnostics exposed a severe cycle-opening asymmetry and loss of the inherited graft capability. No snapshot has yet passed the preregistered canonical-successor leaderboard and capability safeguards, and the already inspected test aggregate is barred from model or threshold selection. Accordingly, **every unconditional, transport, and conditional result below remains a placeholder (XXX)** until either an existing snapshot passes the frozen validation gates or a gated replacement run is trained. The editing run has zero *de novo* mixture weight and does not authorize timed-CTMC generation, so experiment A still requires a separate broad-organic *de novo* checkpoint. Experiment F uses the frozen 967-state `carbon_6_slots` verification slice; its state semantics are fixed, while the exhaustive action audit, kernels, and exact solver results remain pending. No legacy exactness script or stale exactness JSON is used as evidence.

Each experiment tests one link of the causal chain: establish a generative model (A), isolate what the learned rates add (B), test the two capabilities the state space is built for (C, D), check the quotient behaves as proved (E), verify exactness where computable (F), and only then measure the applied regime (G).

[Figure placeholder: panels specified in Appendix A; all values XXX.]

Figure 2: **Generation and trans-dimensional transport** (experiments A–D): unconditional quality against matched baselines, property and size marginals, non-memorization, learned versus uniform legal rewriting, and size and topology adaptation. Panels specified in Appendix A.

What is established before the new experiments. The executor and legal enumerators define a closed bounded state space by construction, and the canonical molecular law is the pushforward of the mark law through that executor. These are implementation and mathematical contracts, not learned-quality results. The completed 16,000-step run establishes that the large packed-data training path is operational, but its mark-level diagnostics also expose unresolved ring-opening and Graft failures. The current program therefore treats ring reachability, finite-budget path cost, teacher-fiber coverage, and learned transport as gated measurements. Their values remain **XXX** until the Active8 corpus, successor-level capacity panels, and corresponding result artifacts are frozen.

Table 2: Experiment design: one row per link in the causal chain, with the comparator that isolates it. No inspected test aggregate is used for model selection, and all learned-result cells remain **XXX** pending the frozen artifacts named in the last column. Full protocols and metric definitions are in Appendix I.

Question	Measured by	Comparator	Status
A Is COMPOSE a generative model?	validity, connectedness, uniqueness, novelty, FCD at matched n , coverage, scaffold novelty, internal diversity, atom-count and cycle-rank distributions, non-memorization	SMILES autoregressive, graph diffusion/flow, insert-delete model, motif model, valid edit-based sampler (Table 6)	XXX ; needs a <i>de novo</i> checkpoint, which does not exist
B What do the learned rates add?	canonical-successor held-out likelihood, analogue recovery, productive displacement, path overhead, reversal and cycle rates, endpoint fidelity	uniform over canonical successors and a state-independent empirical-family law, with identical support, seeds, sources, and budget	XXX ; snapshot selection or a gated replacement run is pending
C Is the process genuinely trans-dimensional?	reaching targets smaller <i>and</i> larger than the source; grow-then-delete inside one trajectory; held-out size bins	no-insert, no-delete, fixed-cardinality, and no-bond_reroute ablations; size-adaptive baselines where runnable	XXX ; the most attackable claim
D Can it redesign ring topology?	cycle creation and opening, fused/spiro/bridged restructuring, cycle-rank change, held-out topology classes	full RingCore vs. no-cycle-operators and no-bond_reroute; finite catalog only if support-honest and runnable	XXX ; the abstract graph mechanism is established below, while chemical reachability remains pending
E Does the quotient do what it claims?	aggregate successor mass vs. sampling frequency; sampled <i>and</i> controlled molecular law under a successor-preserving re-encoding	a deliberately mark-level controller, which must move	XXX ; runnable on any checkpoint
F Is the control exact where computable?	terminal-tilt TV against the analytic target across tilt families; support violations, which must be 0; mid-trajectory switch against restart; value calibration and backward residual	greedy, local Boltzmann, endpoint reranking, rank-directional, particle correction, all on the identical kernel	XXX ; graph rebuilt on the <i>current</i> registry
G Does it help in source-conditioned multi-objective design?	Pareto fronts, normalized HV, HV-AUC against oracle calls, IGD+, feasible nondominated count, source similarity and retention; adaptation regret and cost against restart; the Pareto fan; pathwise feasibility; cardinality and topology correction after a switch; held-out-oracle rescoring	same-base controller/search table including static scalarization and NSGA-II over COMPOSE successors; four frozen dynamic-switch arms; external optimizers for context	XXX ; needs a frozen edit checkpoint

[Figure placeholder: panels specified in Appendix A; all values XXX.]

Figure 3: **Exact and dynamic control** (experiments F–G). *Exact half*: on an enumerable state graph rebuilt from the current operator registry, the exact finite-horizon Doob endpoint law against the analytic target and against greedy, Boltzmann, reranking, rank-directional, and particle controllers on the identical kernel. *Dynamic half*: Pareto fronts and hypervolume against oracle calls, the four registered preference-switch arms (continue from the switch state, restart from the original source, restart from the switch state with a fresh sampler, and a static compromise objective), the Pareto fan, and pathwise constraints. Panels specified in Appendix A.

Why each capability in G is a consequence, not a feature list. Complete intermediates make an oracle query meaningful at every step; birth and death make size adaptation possible; cycle operators make topology redesign possible; the canonical kernel makes successor-level control representation-independent; the remaining-budget value makes retargeting exact; and support restriction makes constraints pathwise. These are causal hypotheses connecting the framework to experiments C–G, not evidence that the corresponding learned capabilities already work. We will compare unguided editing, endpoint reranking, greedy and local-Boltzmann control, MOG-DFM-style guidance where faithfully runnable, SMC, and learned Doob control on one frozen molecular-successor kernel. Superseded checkpoints and pre-RingCore control artifacts are excluded from the evidence ledger rather than used as mechanism results.

8 DISCUSSION AND LIMITATIONS

The contribution is a generative framework whose capabilities are properties of one object: rewriting supplies executable event semantics, Generator Matching learns how probability flows on them, aggregation turns marks into a molecular kernel, and control acts on that kernel. We do not claim to out-optimize strong task-level search: a graph genetic algorithm and PMO-style methods are strong optimizers (Jensen, 2019; Gao et al., 2022), and uniform legal rewriting over our own edit graph is a strong controller the learned rate must beat on the record. Graph validity is closure of the *design space*, not chemistry in a flask: it implies nothing about synthesizability, strain, or oracle reliability, and a rewrite trajectory is not a synthesis route. Exactness is claimed only where h is computable, at the declared horizon, for terminal desirabilities (Remark 8); the deployed controller is an approximation and labelled as one, while support preservation and pathwise invariance carry to scale because they are properties of the support, not of a value function.

Scope boundaries we chose: charge is carried rather than designed; stereochemistry and formal-charge changes are outside the current editing contract; birth has zero or one existing neighbour; and states contain at most 40 active atoms. The bounded Active8 pilot disables both whole-ring growth and whole-ring deletion. Compositional cycle close/open defines graph-topology edits without a finite ring-template catalog, but this does not imply a valid finite-budget path to every chemically labelled ring system. The completed large-data editing run is a diagnostic baseline rather than a selected model: no checkpoint is frozen, ring opening and Graft require successor-level diagnosis, and the editing-V2 rebuild must pass staged capacity and retention gates. Experiment A still requires a separate broad-organic *de novo* checkpoint. Experiment F is being rebuilt on the frozen `carbon_6_slots` verification graph, and no superseded exactness artifact is used as evidence (Appendix I.1). Hard support guarantees imply nothing about FCD, likelihood optimality, mixing, or chemical usefulness.

9 CONCLUSION

We asked whether a generative process can be built out of executable rewrites rather than a generic corruption. Rewrite Generator Matching answers yes, and COMPOSE tests what that construction buys in chemistry: a trans-dimensional process that creates and destroys atoms and changes ring topology, whose every non-null committed state is a supported molecule, whose equivalent mark encodings collapse to one molecular kernel, and whose finite-horizon Doob transform is exact wherever the

bounded kernel is enumerable. COMPOSE learns an executable stochastic rewrite process and turns its induced canonical molecular process into a substrate for exact and dynamic design.

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A FIGURE PLAN

Each main-text figure is specified panel by panel so the plotting scripts can be written against it.

Figure 1 (Figure 1). Rewrite Generator Matching and the trans-dimensional COMPOSE process. Panels: (a) the RGM abstraction: structured state, applicable set $\mathcal{A}(x)$, learned hazard, conditional mark law; (b) the molecular instantiation drawn as strata $\mathcal{X}_n$ ordered by active-atom count, with birth arrows up, death arrows down, and same-cardinality edges inside each stratum, every edge labelled with a public operator name; (c) the Generator-Matching objective over event intensity and complete mark identity and location; (d) canonical successor aggregation, drawn as several visually distinct raw marks converging on one molecular successor, with raw marks and molecules keeping distinct glyphs throughout the paper; (e) compositional ring support: cycle closure and opening build fused, spiro, bridged, and macrocyclic topology with no whole-ring dictionary defining support; (f) the two regimes; (g) the control interface: base kernel, exact or learned value, dynamic goal change, protected subgraph, pathwise constraint; (h) a claim map linking each formal statement to the experiment that verifies it.

Figure 2 (Figure 2). COMPOSE learns a molecular distribution rather than a legal random walk. (a) corpus, source process, trajectory construction, leakage-controlled splits; (b) matched baselines; (c) core distributional metrics over seeds, with dispersion shown; (d) property and size marginals against the held-out reference, including atom-count and cycle-rank distributions; (e) non-memorization; (f) learned transport versus uniform legal rewrites: path length, net displacement, reversals, cycles, family usage, endpoint fidelity; (g) trans-dimensional and topological adaptation: grow-then-delete within one trajectory and held-out ring topologies; (h) scaling against training data, event budget, and model size.

Figure 2 (Figure 2), additional notes. The two adaptation capabilities are the point of the lower half: size adaptation by atom birth and death, and topology change by cycle closure and opening. Show seed dispersion, not means alone.

Figure 3, exact half (Figure 3). Exact finite-horizon control on a bounded molecular kernel. (a) the frozen `carbon_6_slots` verification graph, built from the real executor under its declared bounded semantics; (b) uncontrolled endpoint law and several target tilts; (c) exact Doob endpoint law against the analytic target; (d) greedy, local Boltzmann, endpoint reranking, rank-directional, and particle correction on the identical kernel; (e) support preservation and zero illegal transitions; (f) a dynamic objective switch with the four registered continuation, restart, fresh-sampler, and static-compromise arms, in both law and cost; (g) pathwise support constraints versus terminal filtering;

(h) value calibration and backward-equation residual for the learned controller. The load-bearing result is distributional correctness on the declared slice, not higher reward or representativeness of production chemistry.

Figure 3, dynamic-design half (Figure 3). Support violations must be exactly zero in every panel, and the exact half must show continuation after a mid-trajectory objective change against restarting. The hero trajectory must visibly create atoms, change ring topology, respond to a changed objective by deleting or reorganizing structure it had already built, and finish in a new objective region, with every displayed state a complete molecule.

Source-conditioned multi-objective optimization and dynamic design. (a) benchmark definition: leads, objectives, hard constraints, protected masks, oracle budget, fixed reference point; (b) the same-base controller and search table, including static scalarization and NSGA-II over COMPOSE successors; (c) Pareto fronts, HV, HV-AUC, IGD+, feasible nondominated count, source similarity; (d) external baselines at matched oracle budget; (e) dynamic preference switching across all four registered arms, with adaptation regret and cost; (f) the Pareto fan, one valid prefix branching toward several objective regions; (g) pathwise scaffold, pharmacophore, size, charge, and alert constraints; (h) independent-oracle rescoring with exploitation examples. The hero trajectory must visibly create atoms, change ring topology, respond to a changed objective by deleting or reorganizing structure it had already built, and finish in a new objective region, with every displayed state a complete molecule.

B NOTATION

Symbol	Meaning
$\mathcal{M}_\Gamma = \mathcal{X} = \bigsqcup_{n \leq N_{\max}} \mathcal{X}_n$	state space; $\mathcal{X}_0$ is null and each $\mathcal{X}_n$, $n \geq 1$, contains connected admissible molecular graphs
G, H	states; non-null states are molecules; G_0 source, G_1 data endpoint, G_k^Z program state at progress k
$N_{\max}$	cap on active atoms per state
$a = (r, m, \xi)$	a mark: rule, match, typed payload
$\mathcal{A}(G)$	applicable-mark set at G (state-dependent, guarded)
$T_a G$	executed successor of mark a at G (partial map)
$\mathcal{G}_H(G)$	successor fiber $\{a \in \mathcal{A}(G) : T_a G \simeq H\}$
$\mathcal{S}(G)$	set of distinct successors of G
$\lambda_\theta(G, t)$	learned total hazard (rate, events per unit operational time)
$p_\theta(a G, t)$	learned conditional mark law on $\mathcal{A}(G)$
$q_\theta(a G, t)$	marked rate = $\lambda_\theta p_\theta$
$Q_{\theta, t}(G, H)$	aggregated state-level rate, (4)
$P_\theta^{\text{raw}}(H G, t)$	raw canonical pushforward of the mark law, including self/virtual mass
$\mathcal{S}^+(G), Z_\theta^+(G, t)$	productive successor set and its raw pushforward mass
$P_\theta(H G, t)$	productive canonical-successor kernel for the editing embedded chain
$Q_t(G, H)$	marginal generator of the program mixture (Section 3)
Z, K, K_t	compiled program, its length, latent progress
C, Ψ_t, ω	hard predicate, controller, controller parameters
g, h_b, b, K	terminal desirability, value with b steps left, remaining budget, horizon
G_s, z, m	source lead, preference or target region, protected mask

C EXTENDED RELATED WORK

Generator Matching and jump processes. Generator Matching unifies flow, diffusion, and jump generative models by regressing a marginal generator from tractable conditional generators (Holderrieth et al., 2025); latent-variable formulations extend it to processes with unobserved structure (Billera et al., 2026). RGM is a jump instantiation in which the mark alphabet is state-dependent, guarded, and executable, and in which several marks may map to one successor.

Discrete graph diffusion and flows. DiGress and related discrete diffusion models corrupt node and edge categories and learn a denoiser (Vignac et al., 2023; Austin et al., 2021; Campbell et al.,

Table 3: Positioning ledger. Columns are the properties this paper’s claim rests on; no single column is claimed as new. Var. card.: cardinality changes within one trajectory. Birth/death: entities are both created and destroyed. Topology: operators that change cycle structure. Complete: every visible generative state is a complete molecule. Learned path: a learned reference path distribution rather than a search heuristic. Quotient: probability defined on canonical successors rather than action encodings. Exact ctrl.: exact terminal reweighting with an exact value. Retarget: objective may change mid-trajectory without restarting. Pathwise: constraints enforced at every committed state.

Family	Var. card.	Birth/ death	Topol.	Compl.	Learned path	Quot.	Exact ctrl.	Retgt.	Path- wise
Graph diffusion / flow	✗	✗	✓	✗	✓	✗	✗	△	✗
Trans-dim. jump diff.	✓	✓	△	✗	✓	✗	✗	✗	✗
Insert/delete graph diff.	✓	△	△	✗	✓	✗	✗	✗	✗
Constrained graph diff.	✗	✗	✓	△	✓	✗	✗	✗	△
Molecular grammars	✓	✓	✓	△	✓	✗	✗	✗	△
Valid-state editors	✓	✓	✓	✓	✗	✗	✗	△	✓
Stochastic graph rewriting	✓	✓	✓	✓	✗	△	✗	✗	✓
COMPOSE (this work)	✓	✓	✓	✓	✓	✓	✓	✓	✓

2022); DeFoG formulates graph generation by discrete flow matching (Qin et al., 2025); further variants explore score, spectral, and continuous-time formulations (Jo et al., 2022; 2024; Xu et al., 2024; Siraudin et al., 2025). Their intermediates are discrete graphs but need not be connected or valence-admissible, so an oracle query at an intermediate is not chemically defined.

Variable cardinality and trans-dimensional processes. Trans-dimensional generative modeling via jump diffusion learns processes that move between spaces of different dimension (Campbell et al., 2023); this is the closest formal precedent for (9), and we claim no novelty for variable-dimension generation as such. The distinction is that our dimension changes are typed, guarded graph edits with application conditions, executed by the same executor that validates every other transition, rather than dimension jumps in an ambient continuous space. GrIDDD adds monotone node insertion and deletion to graph diffusion (Ninniri et al., 2025); Edit Flows builds insertion, deletion, and substitution flows for variable-length objects (Havasi et al., 2025); Morph changes cardinality of 3D molecular graphs by unbalanced optimal transport (Franke et al., 2026); DDSBM learns CTMC bridges for graph transformation (Kim et al., 2024).

Grammars, constrained diffusion, and rewriting. Molecular hypergraph grammars make validity structural and improve data efficiency (Kajino, 2019; Guo et al., 2022); fragment and junction-tree models generate through a motif vocabulary (Jin et al., 2018). We claim no novelty for graph grammars. ConStruct’s projector is confined to edge-deletion-invariant topological properties (planarity, acyclicity, lobster structure, bounded degree), whereas a required scaffold or SMARTS motif is edge-insertion-invariant, the opposite class (Madeira et al., 2024); PRODIGY projects onto aggregate count and threshold sets and has no operator for a specific labeled subgraph (Sharma et al., 2024); CoCoGraph fixes formula and degree sequence by double-edge swaps that rewire which atoms bond, so a designated scaffold is not representable (Ruiz-Botella et al., 2026). Stochastic graph rewriting supplies the rule/match/application-condition ontology and CTMC semantics from contextual propensities (Danos et al., 2015; Behr et al., 2021); reaction-based GFlowNets traverse executable synthesis transformations (Kozarski et al., 2024; Zhu et al., 2026).

Editors, optimizers, and controllers. MARS runs annealed MCMC over graph-fragment edits with a learned proposal; MIMOSA iteratively adds, replaces, and deletes under property and similarity constraints; a graph genetic algorithm remains a strong and sample-efficient optimizer, and PMO is the standard efficiency benchmark (Xie et al., 2021; Fu et al., 2021; Jensen, 2019; Gao et al., 2022). These methods couple edit proposals or transition rules to a task-conditioned target distribution or search objective. COMPOSE instead learns an objective-independent stochastic reference transport and applies swappable control at inference; we claim no novelty for valid-state molecular editing itself. Doob h -transforms are classical; inference-time controllers (rank-directional tilting, Feynman–Kac and SMC correctors, value guidance) are orthogonal to RGM and appear here as instances of Ψ in (10).

Table 4: The bounded Active8 development registry used for Gate 0, T1, and P50, classified by its action on the strata of (9). Eight families are active. Whole-ring deletion and growth are disabled. Active8 is a pilot freeze, not a claim that final production support has already been selected. Names in this table are semantic public names; the implementation’s dense family names are engineering aliases.

Public operator	Semantic effect	Kind	Status
atom_insert	create a typed root atom from null, or create one typed atom bonded to exactly one occupied anchor	birth $\mathcal{X}_n \rightarrow \mathcal{X}_{n+1}$	Active8
atom_delete	destroy an eligible atom, leaving a connected non-null remainder, or map a supported one-atom molecule to null	death $\mathcal{X}_n \rightarrow \mathcal{X}_{n-1}$	Active8
atom_restate	jointly change element, valence class, and implicit H at one site	same-cardinality	Active8
bond_reorder	change the order of one existing bond	same-cardinality	Active8
graft	cut a bridge and reattach the detached pendant fragment in one transaction	same-cardinality (reroute)	Active8
cycle_close	insert a bond between two nonbonded atoms, closing a cycle	same-cardinality (cyclize)	Active8
cycle_open	delete a non-bridge cycle edge	same-cardinality (decyclize)	Active8
ring_aromaticity_restate	coordinated joint bond-order restatement of one ring system	same-cardinality	Active8
ring_delete	remove a whole ring system	multi-atom death macro	disabled
ring_grow_macro	grow a whole ring system from a template	multi-atom birth macro	disabled

D OPERATOR SEMANTICS AND CONTRACTS

For each r , $\text{Match}_r(G)$ is the finite set of pattern monomorphisms satisfying structural application conditions. The payload ξ contains new atom/bond states or a macro program. Rule, match, and payload define a partial map $T_{r,m,\xi} : \mathcal{M}_\Gamma \rightarrow \mathcal{M}_\Gamma$. The implementation follows a propose–validate–commit discipline: construct a candidate copy, update implicit hydrogens and bond states jointly, run local valence and global sanitization/connectivity checks, then commit or expose no transition.

Declared scope. States carry at most $N_{\max} = 40$ active heavy atoms, and the heads emit 15 neutral (element, valence) classes over C/N/O/F/S/P/Cl/Br/I/B, a vocabulary admitting 800/800 of the benchmark lead molecules where a neutral C/N/O/F vocabulary admits 33.0%. Formal charge is an input but not a head dimension, so charge is *carried* rather than designed; stereochemistry, isotopes, radicals, and hybridization are not encoded. Appendices D and H give the full ontology, application conditions, charge policy, and scope fingerprint.

Graft. Cut a single bond whose removal would disconnect the graph, choose a reattachment endpoint, and add the new bond in the same atomic transaction, so no disconnected state is ever exposed; atom count and the entire relocated fragment are preserved. In the editing regime the fragment is restricted to a pendant subtree crossing one bond, so ring cores never travel. The successor is identified by the general canonical key, and one shared enumerator serves both the

Table 5: Minimum application conditions for the eight Active8 operators. The production executor imposes the complete grammar-specific checks. Disabled whole-ring growth and deletion are omitted because they contribute no legal marks in the bounded pilot.

Operator	Structural condition	Chemical/transaction condition
atom_insert	empty slot; null source or exactly one occupied anchor	allowed atom state; for connected birth, allowed bond and feasible endpoint valence
atom_delete	removable atom; remainder connected	neighbour H compensation; sanitized successor
atom_restate	occupied slot; successor differs from the source	joint element/valence/H state allowed
bond_reorder	existing edge	joint order/H update feasible
graft	old edge is a bridge; relocated fragment is a pendant subtree; new endpoint outside it	atomic cut+attach; all endpoint valences feasible; successor keyed canonically
cycle_close	absent edge; both endpoints occupied	valence headroom at both endpoints for the new order; connectivity invariant makes the insertion cycle-forming
cycle_open	existing edge that is not a bridge of the current graph	endpoint H compensation; sanitized successor
ring_aromaticity_restate	fixed matched cyclic system	coordinated joint bond-order restatement that changes perceived aromaticity; passes the executor

model’s proposal mask and the teacher construction, which is what guarantees a supervised graft lies inside the mask it is scored against.

Compositional ring construction, in full. `cycle_close` inserts one bond between two non-bonded atoms with valence headroom; because every non-null source is connected, an admitted insertion raises graph cycle rank by one. `cycle_open` deletes one non-bridge cycle edge and lowers graph cycle rank by one. At the untyped graph level, every connected graph is a spanning tree plus its non-tree edges. This explains why primitive close/open operations do not define topology support through a finite ring-template dictionary. It does not prove that the production grammar admits a valence-valid path to every chemically typed ring system, nor that a useful path fits a finite editing budget. Those are executor-replay and experiment D questions. Whole-ring growth and deletion are disabled in Active8. `ring_aromaticity_restate` remains as one coordinated bond-order rewrite, pending its own capacity and reachability gates.

Why coordinated restatement is one rewrite event. Ring-system restatement has one executable successor and one learned event probability. Its payload specifies a coordinated set of bond-order changes that the executor validates as a transaction. Treating such a coupled electronic change as unrelated sequential proposals would expose order-dependent intermediate states and would change the stochastic object. This rationale does not extend automatically to whole-ring addition or deletion, which remain disabled and require separate support, cost, and learnability evidence before any future ablation.

Ring topology coverage, compositionally. Let G be a connected untyped graph with spanning tree T . The non-tree edges $E(G) \setminus E(T)$ number $|E(G)| - |V(G)| + 1$ and determine its graph cycle rank. Inserting those edges one at a time realizes G ’s abstract cyclic topology while preserving connectivity, and removing them in the reverse order restores T . The molecular executor adds typed bond, valence, aromaticity, and support conditions, so the graph construction is a structural argument only. The admissible chemical subset, action-by-action replay success, and finite-budget reachability for monocyclic, fused, spiro, bridged, macrocyclic, and cage targets are measured rather than assumed in experiment D.

E SOURCE PRIORS AND COUPLINGS

E.1 DEGREE-BOUNDED CARBON-TREE SAMPLER

Given size law μ on $\{1, \dots, N_{\max}\}$:

1. sample $N \sim \mu$;
2. sample a Prüfer sequence with rejection or capped construction so every degree is at most four;
3. decode the tree, assign neutral carbon atom states and single bonds, and derive implicit hydrogens;
4. assert connectivity, valence validity, and the declared slot count.

The source sampler consumes only μ and an RNG seed. For a training pair, the tree draw is independent of the endpoint conditional on the selected source size.

E.2 STATIONARY SIZE JITTER

Let J be a Markov kernel on sizes with stationary law μ . Drawing $N_1 \sim \mu$ then $N_0 \sim J(\cdot | N_1)$ preserves the source marginal:

$$\Pr(N_0 = m) = \sum_n \mu(n) J(m | n) = \mu(m). \quad (14)$$

A local proposal $n \rightarrow n \pm 1$ with a Metropolis–Hastings accept/reject correction is a simple reversible choice. The jitter scale controls teacher transport cost but must not be confused with the source law. At inference, N_0 is drawn directly from μ without sampling a target size.

F PROGRAM COMPILATION ALGORITHMS

F.1 PRIMITIVE CONNECTIVITY-PRESERVING TRANSPORT

For completeness, any source can be reduced topologically to one retained atom: delete non-tree edges to expose a spanning tree, then remove leaves. Reverse the construction for a target spanning tree, add chords, and apply typed restatements. The production compiler uses direct common structure and Grafts because the one-atom route is valid but statistically wasteful.

F.2 GRAFT TRANSPORT

Given equal-order carbon tree T_0 and target G_1 :

1. align temporary source slots with target slots and choose a spanning tree $T_1 \subseteq G_1$ maximizing $|E(T_0) \cap E(T_1)|$;
2. while $T \neq T_1$, select $e \in E(T_1) \setminus E(T)$; adding e creates a unique cycle, from which select $f \in E(T) \setminus E(T_1)$; execute the corresponding atomic subtree Graft and validate;
3. restate retained atom states and tree-bond classes in a guard-feasible order;
4. install target chords as cycle-closing bond insertions;
5. canonicalize and assert exact equality with G_1 ; replay the complete program independently.

The graph-level edge-exchange loop performs at most $(n-1) - |E(T_0) \cap E(T_1)|$ iterations. Chemistry guards may reject a nominal exchange, so the implementation searches alternative cycle edges and uses a constructive fallback. A compilation failure is explicit; it is never converted to a truncated target.

F.3 RING MACROS ARE OPTIONAL PROPOSALS, NOT CURRENT SUPPORT

The bounded Active8 registry uses no whole-ring growth or deletion event. Ring topology is edited through primitive cycle close/open operations, while coordinated ring-system restatement handles

admitted bond-order changes. A future ring macro would alter one-event reachability and path cost under a fixed budget, even if its lowering used only primitive operations. It must therefore be treated as a registered efficiency proposal with matched primitive-equivalent accounting, not assumed to preserve the finite-horizon process.

Any future macro catalog must be frozen before evaluation and audited for the observed transformations it covers, the tail it excludes, executor replay, canonical-successor aliasing, and capacity under matched compute. A finite-catalog comparator belongs in experiment D only if it is legitimately runnable and support differences are reported. It does not define the current framework or rescue a failed primitive operator by assumption.

F.4 PROGRAM-CACHE REPRODUCIBILITY

Programs are stored as rewrite instructions plus byte-packed exact graph checkpoints rather than a dense graph at every progress position. A sampled state replays at most $c - 1$ validated actions from the nearest checkpoint. CPU compilation writes atomic, signature-validated shards with a manifest containing split hashes, compiler version, chemistry grammar, operator-capability and support hashes, prior or coupling configuration, seeds, and checkpoint interval. A GPU training stage refuses to start from an incomplete cache. These choices change throughput and storage, not the mathematical path distribution.

G PROOFS

G.1 PROOF OF PROPOSITION 3 (CLOSURE HALF)

Induct on committed events. The base state is in $\mathcal{M}_\Gamma$. If $G_n \in \mathcal{M}_\Gamma$, the CTMC has support only on $a \in \mathcal{A}(G_n)$; by (1), $G_{n+1} = T_a G_n \in \mathcal{M}_\Gamma$. Local boundedness excludes explosion on the finite horizon, so only finitely many jumps occur almost surely. The conclusion applies equally to teacher programs and learned sampling because both invoke the same executor. $\square$

G.2 PROOF OF PROPOSITION 6 (CLOSURE UNDER CONTROL, AND PATHWISE INVARIANCE)

The controlled off-diagonal rate (10) is a pointwise product of a nonnegative indicator, the learned rate, and a nonnegative controller, hence nonnegative, and it vanishes whenever $Q_{\theta,t}(G, H) = 0$. By (4), $Q_{\theta,t}(G, H) > 0$ only if some $a \in \mathcal{A}(G)$ has $T_a G \simeq H$, and then $H \in \mathcal{M}_\Gamma$ by (1). So the controlled process has support contained in $\mathcal{S}(G) \subseteq \mathcal{M}_\Gamma$, and the closure induction above applies verbatim with $Q_{\theta,t}$ replaced by $\tilde{Q}_t$. Local boundedness is preserved because the hypothesis gives a finite $\sup \Psi_t$ on base-supported transitions over the registered finite horizon, hence $\tilde{Q}_t \leq (\sup \Psi_t) Q_{\theta,t}$. If $C \equiv 1$ and $\Psi_t(G, H) > 0$ at every base-supported $H \neq G$, then $\tilde{Q}_t(G, H) > 0$ iff $Q_{\theta,t}(G, H) > 0$ off the diagonal, so those supports coincide. Neither factor can make $\tilde{Q}_t(G, H) > 0$ where $Q_{\theta,t}(G, H) = 0$, which is the assertion that no controller creates support. $\square$

Pathwise invariance. For the second claim, let $\mathcal{A}_P(G) = \{a \in \mathcal{A}(G) : P(T_a G)\}$ be the restricted applicable set, i.e. (10) with $C = P$ and $\Psi \equiv 1$. Induct on committed events. $P(G_0)$ holds by hypothesis. If $P(G_k)$ holds, the controlled process commits only $a \in \mathcal{A}_P(G_k)$, and $P(T_a G_k) = P(G_{k+1})$ holds by the definition of $\mathcal{A}_P$. Hence $P(G_k)$ for every committed k , almost surely. Feasibility is decided before commitment. When P is separate from a downstream scoring oracle, rejected successors can be discarded before that oracle is called. $\square$

Remark 9. The predicate must be evaluable on a complete molecule, which is exactly what the state space guarantees and what an ambient-space process does not: for an incomplete or possibly disconnected intermediate, "will this trajectory end containing a specified labeled substructure" is not decidable from the intermediate, so the restriction $\mathcal{A}_P$ cannot be formed.

G.3 PROOF OF PROPOSITION 5 (ENCODING INVARIANCE)

(i) By Definition 4 the fibers correspond: φ_G maps $\mathcal{G}'_H(G)$ into $\mathcal{G}_H(G)$ for every H , and surjectivity plus successor preservation make $\{\mathcal{G}'_H(G)\}_H$ a partition of $\mathcal{A}'(G)$ refining nothing across fibers.

Rate preservation then gives, for every G, H, t ,

$$Q'_{\theta,t}(G, H) = \sum_{a' \in \mathcal{G}_H(G)} q'(a' | G, t) = \sum_{a \in \mathcal{G}_H(G)} q(a | G, t) = Q_{\theta,t}(G, H).$$

Summing over H yields $\lambda'_\theta(G, t) = \sum_H Q'_{\theta,t}(G, H) = \lambda_\theta(G, t)$, so the total hazards agree without any further hypothesis. This is the answer to the natural worry that a refinement, which must renormalize p_θ over a larger applicable set, could perturb the hazard: it cannot, because the hazard is the sum of the aggregates. The generator $(L_{\theta,t}f)(G) = \sum_{H \neq G} Q_{\theta,t}(G, H)[f(H) - f(G)]$ therefore agrees, and by uniqueness of the finite-state Kolmogorov forward solution under assumption 1 so do all time marginals and the terminal law. On every row with positive total hazard, $P_\theta^{\text{raw}}(H | G, t) = Q_{\theta,t}(G, H)/\lambda_\theta(G, t)$, so the raw pushforwards agree. Because the productive predicate is the same state-level predicate in both encodings, (5) then gives $P_\theta = P'_\theta$ on every row with positive productive mass; the same declared absorbing convention handles stalled rows. Thus the productive embedded jump chains agree as well. Note this is an identity of laws over *state* trajectories; the two systems obviously induce different laws over *mark* trajectories, which are over different alphabets.

(ii) If Ψ_t is fiber-constant then it is a function $\Psi_t(G, H; \omega)$, so

$$\begin{aligned} \tilde{Q}'_t(G, H) &= \mathbf{1}\{C(H)\} Q'_{\theta,t}(G, H) \Psi_t(G, H; \omega) \\ &= \mathbf{1}\{C(H)\} Q_{\theta,t}(G, H) \Psi_t(G, H; \omega) = \tilde{Q}_t(G, H), \end{aligned}$$

and the diagonal, being determined by the off-diagonal row, agrees too. The hard mask is fiber-constant by definition, since C is a predicate on states.

(iii) h_b is defined by backward recursion from the productive kernel at each remaining budget, every one of which is invariant by (i), so $h_b = h'_b$ for every b by induction. The two encodings therefore have the same reachable set $\{G : h_b(G) > 0\}$, and on that set $\Psi_b(G, H) = h_{b-1}(H)/h_b(G)$ is invariant and fiber-constant. The controlled kernels of Theorem 7 agree wherever they are defined. $\square$

Adversarial checks that the statement must survive, and one it does not. Four attack surfaces were tested before this proposition was stated. (1) *Hazard renormalization*: handled in (i), the hazard is the sum of the aggregates, so preserving all aggregates preserves it, and no extra hypothesis is needed. (2) *Merging marks with different legality masks*: if a_1, a_2 have the same successor but differ in legality at G , only the legal ones are in $\mathcal{A}(G)$, so the aggregate is over the legal subset and the hypothesis is stated on $\mathcal{A}(G)$; a merge is admissible exactly when the merged mark’s legality reproduces that aggregate. What is *not* admissible is a merge across successors, which would make T ill-defined on the merged system, and Definition 4 excludes it, and the informal “any refinement or coarsening” phrasing does not, which is why the condition is stated. (3) *Composition with the Doob transform*: see (iii); it survives, because the value is a function of the invariant kernel. (4) *Refinements that change the applicable-set cardinality*: these are the generic case and are covered: $|\mathcal{A}'(G)| \neq |\mathcal{A}(G)|$ is expected under splitting, and nothing in the argument uses the cardinality, only the fiber sums. *The check it fails.* A controller that is not fiber-constant breaks (ii), and this is not a corner case. Let $\mathcal{G}_H(G)$ contain four symmetric aliases of rate $1/4$ under one encoding and one mark of rate 1 under another, with all other successors identical. A mark-level power tilt $\Psi(a) \propto q(a)^{\beta-1}$ assigns H controlled mass $4 \cdot (1/4)^\beta$ versus 1, which differ for every $\beta \neq 1$. Top- k or nucleus truncation over marks and per-family tilts therefore generally fail; a family multiplier is an exception exactly when it is constant across all aliases of every successor fiber it touches. Two further scope limits are recorded rather than hidden: the proposition is about the *population* process, so it says nothing about gradients, variance, or sample efficiency; and the training *objective* is invariant only when the teacher term is aggregated, since $r \log q_\theta(a)$ evaluated at one alias changes under splitting while $r \log \sum_{a \in \mathcal{G}_H} q_\theta(a)$ does not. This is precisely why the graft teacher in the historical run was scored by a log-sum-exp over its successor group and why (8) aggregates every teacher fiber.

G.4 PROOF OF THEOREM 7

Write $\Pr_x$ for the law of the uncontrolled budget-indexed chain started at x and $h_b(x) = \mathbb{E}_x[g(X_b)]$, which satisfies $h_0 = g$ and the stated backward recursion by conditioning on the first step. On a row with $h_b(x) > 0$, (11) is well defined and stochastic because $\sum_y P_{\theta,b}(x, y) h_{b-1}(y) = h_b(x)$. Its support is contained in the corresponding base row because multiplication by $h_{b-1}(y)$ cannot

make a zero base probability positive. The two supports are equal exactly when every base-supported successor has positive h_{b-1} . If $h_b(x) = 0$, no path from x reaches a positive-desirability endpoint in b steps, so x is unreachable under the conditioned law and no transformed row is defined there.

For a path $x_0, \dots, x_K$, write

$$\pi_K(x_{0:K}) = \prod_{k=0}^{K-1} P_{\theta, K-k}(x_k, x_{k+1}).$$

If $\pi_K(x_{0:K}) > 0$, then under the controlled chain, whose step from x_k uses budget $b = K - k$,

$$\begin{aligned} \Pr^g(X_{0:K} = x_{0:K}) &= \pi_K(x_{0:K}) \prod_{k=0}^{K-1} \frac{h_{K-k-1}(x_{k+1})}{h_{K-k}(x_k)} \\ &= \pi_K(x_{0:K}) \frac{h_0(x_K)}{h_K(x_0)}, \end{aligned}$$

because the h factors telescope. Since $h_0 = g$, every path to a fixed endpoint y is reweighted by the same factor $g(y)/h_K(x_0)$, and summing over paths gives (12). The normalizer is $h_K(x_0) = \mathbb{E}_{x_0}[g(X_K)]$, the value with the *full* budget at the start.

For the continuation statement, apply the same argument to the chain started at x with horizon b' and terminal desirability g' : the telescoping is over $k = 0, \dots, b' - 1$ with $h'_{b'}$ as normalizer, so the endpoint law is the g' -tilt of $\Pr_x(X_{b'} = \cdot)$. Note this is in general *not* the same as having used g' from step 0, which is why comparing continuation against restart is a meaningful experiment rather than a tautology.

Finally, the marginal at $j < K$ is $\Pr^g(X_j = y) = \Pr(X_j = y \mid x_0) h_{K-j}(y)/h_K(x_0)$, a h_{K-j} -tilt rather than a g -tilt; these coincide only when $j = K$. Hence early stopping does not inherit (12), as stated in Remark 8. $\square$

G.5 PROOF OF PROPOSITION 3 (REACHABILITY HALF)

For connected G , choose a spanning tree T_G . Delete every edge in $E(G) \setminus E(T_G)$; each is on a cycle at deletion and is therefore not a bridge, so connectivity is preserved. Repeatedly delete leaves until one vertex remains. From one vertex, reverse leaf insertions to construct a spanning tree T_H of H , then insert edges in $E(H) \setminus E(T_H)$. Every intermediate is connected.

For aligned trees T and T_1 , take $e \in E(T_1) \setminus E(T)$. Adding e creates a unique cycle. That cycle contains an edge $f \in E(T) \setminus E(T_1)$; otherwise T_1 would contain a cycle. Replacing f by e is one subtree Graft, preserves the tree property, and increases overlap with T_1 by one. Repetition terminates after at most $(n - 1) - |E(T_0) \cap E(T_1)|$ exchanges. $\square$

G.6 PROOF OF PROPOSITION 2 (POPULATION MINIMIZER)

Fix occupied (G, t, H) and let $Y = R_t^Z(H \mid X_t, K_t)$ conditional on $X_t = G$. The part of (6) depending on scalar prediction $u \geq 0$ is

$$\ell(u) = u - \mathbb{E}[Y \mid X_t = G] \log u, \quad (15)$$

with target-only terms omitted. For $u > 0$, $\ell'(u) = 1 - \mathbb{E}[Y \mid X_t = G]/u$ and $\ell''(u) = \mathbb{E}[Y \mid X_t = G]/u^2 \geq 0$. If the mean is positive, the unique minimizer is its value; if zero, the infimum is attained at $u = 0$ by continuity. Finite sums commute with conditional expectation, so the result applies after alias aggregation. $\square$

G.7 PROOF OF PROPOSITION 2 (ENDPOINT LAW)

The program mixture induces marginal law p_t satisfying the Kolmogorov forward equation with the marginal generator Q_t of Section 3; this is the standard conditional-to-marginal Generator Matching identity. Under assumption 1, the finite-state forward equation has a unique solution. If $Q_{\theta, t} = Q_t$ and both laws begin at π_0 , their marginals agree for all t . By assumption 1, the program mixture terminates at p_{data} , hence so does the exact learned process. $\square$

G.8 IMPORTANCE-CORRECT PROGRESS STRATIFICATION

Let $p(k | t, Z)$ be the binomial progress law and $h(k | Z)$ a family-balanced proposal. Sample from $\tilde{p} = (1 - \rho)p + \rho h$ and weight by $w = p/\tilde{p}$. For any integrable per-progress loss g , $\mathbb{E}_{\tilde{p}}[wg] = \sum_k p(k)g(k) = \mathbb{E}_p[g]$. Since $\tilde{p} \geq (1 - \rho)p$, support is retained for $\rho < 1$.

H MODEL, TRAINING, AND SAMPLING DETAILS

H.1 NEURAL ARCHITECTURE

The graph encoder embeds atom type, formal charge, implicit hydrogen count, bond class, model time, and ring/topology context. Message passing produces node and graph representations. A graph head predicts total hazard where the timed process requires it; masked family and operand heads score complete legal marks. Canonical-successor training sums the probability of all marks that execute to the teacher molecule, rather than rewarding one arbitrary alias. The current candidate backbone has hidden width 256 and 6 message-passing layers, but the released architecture, parameter count, optimizer, learning-rate schedule, precision, batch size, throughput, and peak memory remain **XXX** until the T1 and P50 gates freeze them. Scratch initialization is allowed. If a warm start is used, semantically matching parameters are transferred by label and inherited capabilities receive explicit parity and retention probes; no de novo checkpoint is required by the framework.

H.2 TRAINING ESTIMATOR

For each minibatch item:

1. draw or load (G_0, G_1, Z) from the declared coupling and compiled cache;
2. sample time t and progress k , optionally from the importance-correct stratified proposal;
3. reconstruct $X_t = G_k^Z$ from its nearest packed checkpoint;
4. encode X_t once and form analytic legality masks;
5. evaluate the objective bound to the run contract: (8) for the productive editing-V2 lane, a calibrated rate objective for the timed *de novo* lane, or (6) only in an implementation that actually parameterizes its aggregated rate terms;
6. aggregate every alias of the teacher successor before the editing identity loss and update θ .

We log hazard separately from productive canonical-successor NLL, teacher-successor probability and rank, raw mark count, canonical successor count, alias multiplicity, effective update mass, family/operand diagnostics, illegal-mask disagreements, and gradient norm. Model selection uses the frozen validation-only canonical-successor criterion and safeguards, never test FCD or the already inspected historical test aggregate.

H.3 PIECEWISE-CONSTANT TIMED AND EMBEDDED-CHAIN SAMPLERS

Let $0 = s_0 < \dots < s_J = S$ be the operational-time grid and define the registered de novo rate model to be piecewise constant in operational time on every interval $[s_j, s_{j+1})$ (while still recomputing after a state change). Starting from $G \sim \pi_0$, at grid interval j :

1. compute $\lambda = \lambda_\theta(G, s)$ and masked marked probabilities;
2. if $\lambda = 0$, advance to s_{j+1} and log a stall;
3. otherwise draw $\Delta \sim \text{Exponential}(\lambda)$;
4. if $s + \Delta \geq s_{j+1}$, set $s = s_{j+1}$ without an event;
5. else sample a hierarchically, execute $G \leftarrow T_a G$, set $s \leftarrow s + \Delta$, and recompute all rates.

This is exact Gillespie simulation of the declared piecewise-constant time-inhomogeneous CTMC. If instead the network is interpreted as varying continuously with time, left-boundary freezing is an approximation, not an exact ancestral sampler, and grid-refinement convergence and endpoint

(a) Timed CTMC (*de novo*). Sample $G \sim \pi_0$, set operational time $s = 0$. Under the registered piecewise-constant time schedule, hold rates fixed between time-grid boundaries. While $s < S$ and the event budget is not exhausted: (i) predict the total legal hazard and masked mark probabilities; (ii) draw an exponential waiting time; (iii) if it crosses the next boundary, advance to that boundary, otherwise sample one legal marked rewrite, execute $G \leftarrow T_a G$, and advance by the waiting time.

(b) Embedded jump chain (editing). Sample $G = G_s$ (the lead), set the remaining budget $b = K$. While $b > 0$ and $Z_\theta^+(G, t) > 0$: sample a productive successor $H \sim P_\theta(\cdot \mid G, t)$ under the successor-level controller of Section 6; then sample an executable alias $a \in \mathcal{G}_H(G)$ from $p_\theta(a \mid G, t) / P_\theta^{\text{raw}}(H \mid G, t)$ and execute $G \leftarrow T_a G \simeq H$; decrement b . Return the current molecule, or the archive of all visited molecules.

Figure 4: The two inference regimes share a rate-law factorization and executor but use separately fitted checkpoints. The *de novo* timed CTMC uses the hazard for event timing; source-conditioned editing uses the productive canonical-successor jump chain at a fixed edit budget.

sensitivity must be reported. The event budget is a numerical safeguard and its hit rate is reported. There is no rejection loop over invalid actions: analytic masks and the executor define zero support, and any disagreement is an auditable implementation error.

H.4 COMPLEXITY

Let $n \leq N_{\text{max}}$, d be hidden width, L message-passing layers, and $C_r(G)$ the tensorized candidate count for family r . One rate evaluation costs the graph encoder plus masked heads, approximately $O(L(nd^2 + |E|d) + \sum_r C_r(G)d)$ for the dense implementation, rather than one graph encoding per canonical successor. Compilation is offline and program-dependent; Graft tree alignment is polynomial and each edge exchange is validated. Final measured preprocessing time, peak memory, train throughput, and sample throughput are XXX, XXX, XXX, and XXX.

I COMPLETE EXPERIMENTAL PROTOCOL

I.1 EMPIRICAL STATUS, AND WHAT MAY BE CLAIMED

We record the state of the empirical record precisely, because it bounds every claim in Section 7.

Completed editing run. The first large-data scientific run completed 16,000 steps over 661,105 admitted training traces and produced all 32 scheduled snapshots. It is an informative diagnostic baseline, not a selected production checkpoint. Its training objective was mark-level Generator Matching. Mark-level held-out diagnostics show that ring closing learned while ring opening remained poor, and that an inherited Graft capability collapsed during training. Canonical-successor evaluation across all snapshots, artifact hashing, family trajectories, and forensic support audits remain prerequisites to deciding whether any checkpoint is usable.

Editing-V2 rebuild. The next editing model uses a content-addressed corpus with frozen structured roles, exact slot-addressed states, whole-trace Active8 admission, and a canonical-successor objective for the productive embedded jump chain. Training proceeds only through Gate 0, T1, P50, P500, and P2000 decision artifacts. Scratch initialization is permitted. A warm start, if used, must pass initialization-parity and retention probes. Until those gates pass and a validation-only checkpoint is frozen, all learned transport, adaptation, and controlled-design cells remain XXX.

Separate *de novo* and exact-control lanes. The editing checkpoint does not support the unconditional claim. Experiment A requires its own broad-organic checkpoint, nonzero *de novo* training mixture, source distribution, and calibrated hazard. Experiment F uses the separately frozen `carbon_6_slots` graph, which currently contains 967 states and 14,432 canonical directed edges. That carbon-only graph is a bounded algebraic verification slice, not representative chemistry and not the production vocabulary. Its exhaustive action classification, kernel construction, exact backward solver, and learned-checkpoint projection are pending.

Table 6: **Unconditional molecular generation** (experiment A). Training corpus, parameter count, and accelerator hours are disclosed per row. Every non-null committed COMPOSE state must be valid and connected by construction (Proposition 3); endpoint validity, all-step non-null validity, connectedness, and null-endpoint rate are reported separately, and an invalid non-null state is a software failure. Desc. W_1 is the mean physicochemical descriptor Wasserstein distance; FCD is computed at matched sample count with the reference reported (Appendix I). All COMPOSE cells are **XXX** pending a *de novo* checkpoint and a frozen multi-seed evaluation (**XXX** seeds, **XXX** samples).

Method	Valid $\uparrow$	Conn. $\uparrow$	Unique $\uparrow$	Novel $\uparrow$	FCD $\downarrow$	Desc. W_1 $\downarrow$	Int. div. $\uparrow$	Cov. $\uparrow$
SMILES LSTM (Brown et al., 2019)	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX
DiGress	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX
DeFoG	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX
GrIDDD	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX
Morph	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX
Fragment / hierarchy (Jin et al., 2018; Lee et al., 2026)	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX
Valid edit-based sampler	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX
COMPOSE	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX

I.2 FCD HYGIENE

FCD is computed with one frozen ChemNet implementation and environment; invalid or null samples are not replaced. We report generated and reference counts, total FCD, the declared decomposition diagnostics, multi-seed mean and standard deviation, and bootstrap intervals. Every comparison uses matched sample counts and reports the corresponding reference value. Cross-paper FCD values are contextual only when sample count, implementation, and reference distribution differ.

I.3 EXACT-CONTROL VERIFICATION PROTOCOL

The `carbon_6_slots` state graph is indexed deterministically from the current persistent-slot semantics and real executor. Every raw mark is classified as an included edge, an excluded boundary action, a self or virtual event, or an explicit failure. A slow dictionary pushforward independently groups executed marks by canonical molecular successor and must agree with the production segmented implementation. The resulting budget-indexed kernels must be row stochastic. Backward values, transformed kernels, terminal laws, support restriction, unreachable $h_b(x) = 0$ states, and dynamic continuation are then checked against explicit path enumeration on the smallest tractable sub-slice. All numeric cells remain **XXX** until the graph audit and solver artifacts pass.

I.4 HISTORICAL EVIDENCE EXCLUSIONS

Earlier exact-Doob scripts, exactness JSON files, cap-four or cap-five task selections, and figures built from a pre-RingCore registry are developmental artifacts and are not evidence for experiment F. Results from the superseded CNOF *de novo* base and data-starved editing checkpoints are likewise excluded from experiments A–G. They may be used only to identify software paths or motivate preregistered tests. No number from those artifacts is promoted into the abstract, results, figures, tables, checkpoint selection, threshold setting, or causal comparison.

I.5 PENDING RESULT TABLES

Non-memorization. A generator over a legal edit graph could in principle reproduce training molecules by transporting to memorized endpoints. We therefore report nearest-training-neighbour similarity distributions, duplicate growth as a function of sample count, and scaffold overlap with the training split (Figure 2e), and we hold out whole scaffold series so that “novel” means novel with respect to a series rather than a single molecule.

I.6 DEVELOPMENTAL EVIDENCE IS NOT A BENCHMARK RESULT

Developmental runs may establish that a code path executes, but they do not establish learned quality. Earlier size-matched Graft diagnostics, restricted-chemistry rollout counts, data-starved checkpoints, and pre-RingCore control artifacts are excluded from the scientific evidence ledger. This draft retains

Table 7: **Same-base controller and search comparison** (experiment G). Every row uses the same frozen learned molecular kernel or its declared canonical-successor support, operator system, source set, edit budget **XXX**, oracle budget **XXX**, and reference point **XXX**. Endpoint reranking is the same-generator post-hoc arm; NSGA-II searches over COMPOSE successors rather than changing the executor. Paired over identical sources, ≥ 5 repetitions, bootstrap intervals. All values **XXX**.

Controller	Norm. HV $\uparrow$	HV-AUC $\uparrow$	IGD+ $\downarrow$	Feasible ND $\uparrow$	Source sim.
Unguided (learned prior only)	XXX	XXX	XXX	XXX	XXX
Uniform legal rewrites	XXX	XXX	XXX	XXX	XXX
Endpoint reranking	XXX	XXX	XXX	XXX	XXX
Greedy local improvement	XXX	XXX	XXX	XXX	XXX
Local Boltzmann tilt	XXX	XXX	XXX	XXX	XXX
Static scalarized guidance	XXX	XXX	XXX	XXX	XXX
Rank-directional (MOG-style)	XXX	XXX	XXX	XXX	XXX
SMC / Feynman-Kac	XXX	XXX	XXX	XXX	XXX
Learned finite-horizon Doob (13)	XXX	XXX	XXX	XXX	XXX
NSGA-II over COMPOSE successors	XXX	XXX	XXX	XXX	XXX

Table 8: **Main table 3: external molecular-optimization baselines at matched oracle budget.** Context, not a causal comparison: these differ in pretraining, representation, support, and search. A row runs only after its registered adapter and task-compatibility gates pass. Oracle calls count *all* evaluated intermediates and proposals. Published numbers are not substituted for a mismatched task. All values **XXX**.

Method	Norm. HV $\uparrow$	HV-AUC $\uparrow$	Feasible ND $\uparrow$	Source sim.	All-state valid
Graph GA (Jensen, 2019)	XXX	XXX	XXX	XXX	XXX
MARS (Xie et al., 2021)	XXX	XXX	XXX	XXX	XXX
RetMol (Wang et al., 2023)	XXX	XXX	XXX	XXX	XXX
GenMol (Lee et al., 2025)	XXX	XXX	XXX	XXX	XXX
InVirtuoGen (Kaeche et al., 2026)	XXX	XXX	XXX	XXX	XXX
HN-GFN (Zhu et al., 2023)	XXX	XXX	XXX	XXX	XXX
COMPOSE (learned Doob)	XXX	XXX	XXX	XXX	XXX

Table 9: Ring-taxonomy and size marginals against a held-out reference. All values **XXX** pending the frozen evaluation. The *de novo* base checkpoint’s known small-ring/aromaticity defect is the motivation for the compositional ring operators, and this table is the test of whether they fix it.

	Small ring (3–4)	Aromatic frac.	Fused	Bridged	Spiro	Macrocycle
Reference	XXX	XXX	XXX	XXX	XXX	XXX
COMPOSE	XXX	XXX	XXX	XXX	XXX	XXX

no numerical result from them. New engineering gates write content-addressed artifacts and are reported only as gate outcomes until the corresponding frozen experiment is run.

I.7 FROZEN DATA CONTRACT

We will release raw input hashes, canonicalization code, exact slot-addressed states, molecular identity keys, split assignments, $N_{\max}$, element, charge and bond vocabularies, and every exclusion ledger. Splits are frozen before reusable components, transformation statistics, templates, or learned preprocessing are derived. Validation controls model selection; controller-validation controls controller choices; the sealed final test influences neither.

I.8 FCD CONTRACT

FCD is computed with one frozen ChemNet implementation and environment. Invalid or null samples are not replaced. We report generated/reference counts, total FCD, additive mean and covariance terms, generated/reference covariance traces, multi-seed mean and standard deviation, and bootstrap intervals. We use at least **XXX** independent samples per final seed; any 2,000-sample diagnostic is labeled preliminary.

Table 10: Matched ablations isolating which construction choices produce the observed behaviour. Teacher path length is measured on compiled programs; distribution distance and ring-size TV are measured from target-free learned samples. Ring-macro or finite-catalog rows run only if their registered implementation and support-honesty gates pass. All values **XXX** pending the frozen study.

Variant	Desc. $W_1 \downarrow$	Ring-size TV $\downarrow$	Mean teacher length $\downarrow$	Endpoint fidelity $\uparrow$
Uniform legal rates (no learning)	XXX	XXX	—	XXX
No atom insertion	XXX	XXX	XXX	XXX
No atom deletion	XXX	XXX	XXX	XXX
Fixed cardinality	XXX	XXX	XXX	XXX
No <code>bond_reroute</code>	XXX	XXX	XXX	XXX
No cycle operations	XXX	XXX	XXX	XXX
Null source	XXX	XXX	XXX	XXX
Primitive tree (delete/regrow)	XXX	XXX	XXX	XXX
Tree + graft	XXX	XXX	XXX	XXX
Compositional rings only	XXX	XXX	XXX	XXX
Ring catalog only	XXX	XXX	XXX	XXX
Compositional + ring macros	XXX	XXX	XXX	XXX
No successor alias aggregation	XXX	XXX	XXX	XXX
No importance-correct stratification	XXX	XXX	XXX	XXX
COMPOSE (full)	XXX	XXX	XXX	XXX

Table 11: Path and structural fidelity across source priors and couplings. All-step metrics audit every accepted intermediate, not only endpoints. Values pending confirmatory evaluation.

Variant	Step valid	Step conn.	Size TV	Cycle TV	Ring TV	Budget	Events
Null prior	XXX	XXX	XXX	XXX	XXX	XXX	XXX
Independent tree	XXX	XXX	XXX	XXX	XXX	XXX	XXX
Size-matched graft	XXX	XXX	XXX	XXX	XXX	XXX	XXX
Flexible-size graft	XXX	XXX	XXX	XXX	XXX	XXX	XXX

Table 12: Controllers on the *identical* enumerated molecular kernel, scored by total variation to the exact terminal tilt (lower is better) rather than by attained reward. Support violations must be exactly zero for every row, since no controller in (10) can create support. All values **XXX** pending the frozen enumerable-space study.

Controller	TV to exact tilt $\downarrow$	Target-event prob. err. $\downarrow$	Support viol.	Switch regret $\downarrow$
Unguided reference	XXX	XXX	XXX	—
Greedy local reward	XXX	XXX	XXX	XXX
Local Boltzmann guidance	XXX	XXX	XXX	XXX
Endpoint reranking	XXX	XXX	XXX	XXX
MOG-style rank-directional	XXX	XXX	XXX	XXX
Particle correction (SMC/FK)	XXX	XXX	XXX	XXX
Exact finite-horizon Doob	XXX	XXX	XXX	XXX
Learned controller (approximation)	XXX	XXX	XXX	XXX

I.9 ABLATION PROTOCOLS

Prior/coupling. Hold architecture, data, steps, and sampler fixed; compare null, independently sized carbon tree, size-matched tree, and stationary size-jitter. Report teacher length as well as learned endpoint metrics.

Graft. Compare primitive delete/regrow transport with Graft-enabled transport on identical sampled source–target pairs. Measure compilation success, path length, one-atom visits, source-edge survival, training throughput, event use, and endpoint FCD/topology.

Rings. Compare scalar bond construction, topology-only ears, typed ears, and typed ears plus ring-system restatement. Report per-class recall/TV for aromatic, saturated, fused, bridged, spiro, small rings (3–4), ordinary rings (5–7), and macrocycles under the dataset definition.

Objective and quotient. Ablate total-hazard learning, importance-correct stratification, and canonical alias aggregation. The last ablation tests whether symmetric syntactic multiplicity biases chemical successor rates.

I.10 TRAJECTORY AUDITS

For every sampled event we record time, rule family, operands, pre/post canonical keys, hazard, and validity/connectivity flags. Aggregates include:

- rule-family and inverse-pair utilization by time and size;
- source-edge survival, topology edit distance, and one-atom-state visitation;
- ring class before and after every cyclic event;
- aliases per successor and marked-versus-quotient calibration;
- event-budget exhaustion, zero-hazard stalls, and time-grid sensitivity;
- agreement of analytic masks/direct sampling with exhaustive fibers on small graphs.

I.11 ADDITIONAL PENDING RESULTS

Table 13: Ring taxonomy distribution distances; all values pending.

Method	Aromatic	Saturated	Fused	Bridged	Spiro	Macrocycle
DiGress	XXX	XXX	XXX	XXX	XXX	XXX
Morph	XXX	XXX	XXX	XXX	XXX	XXX
COMPOSE	XXX	XXX	XXX	XXX	XXX	XXX

Table 14: Compute and storage; preprocessing is included and all values are pending.

Method	Parameters	Compile h	Train h	Peak memory	Molecules/s
COMPOSE	XXX	XXX	XXX	XXX	XXX

I.12 CONTROLLED-DESIGN AND EDITING PROTOCOL

Source-conditioned controlled design is the paper’s applied regime (experiment G). Sources come from the frozen broad-organic editing evaluation roles, never the superseded CNOF fixture. A registered scaffold, pharmacophore, atom-count, charge-policy, structural-alert, or optional similarity-corridor predicate may remove incompatible successors before commitment. We will report completely feasible trajectory rate, endpoint yield, generate-then-filter usable fraction by hardness, usable-oracle efficiency, constrained property gain, scaffold retention, anytime best-so-far reward, edit count, diversity, all-step validity, and every failure or abstention. Every value remains **XXX** until the editing checkpoint, tasks, objectives, normalization, hypervolume references, budgets, seeds, and independent evaluator are frozen and the registered run is complete. Reward fine-tuning, inference-time guidance, and the unguided prior are reported as distinct arms.

Dynamic preference switching. The switch step and both objectives are frozen before results. Four arms use matched accounting: *continue from the current switch state* with the remaining budget; *restart from the original source*; *restart from the switch-state molecule with a fresh sampler*; and a *static compromise objective* fixed from step zero. We report adaptation regret, retained gain on the old objective, final gain on the new objective, edit and oracle cost, and time to adapt. No two-arm “continue versus restart” summary substitutes for this registered comparison.

J FAILURE MODES, BROADER IMPACT, AND REPRODUCIBILITY

Failure modes. We stratify failures into compiler support, proposal-mask false negatives, executor discrepancies, zero-hazard stalls, event-budget exhaustion, source-memory artifacts, invalid ring

electronics, over-fusion, macrocycle excess, and endpoint distribution mismatch. Chemical examples are selected by predeclared strata rather than only visual appeal.

Broader impact. The model can generate valid but unsafe, toxic, or controlled compounds. A distribution-learning score is not evidence of therapeutic value. Downstream property guidance requires access control, dual-use review, toxicity assessment, and wet-lab governance. Corpus bias and cheminformatics blind spots remain even when formal validity is guaranteed.

Artifact checklist. The final release will contain:

- code version, environment lockfile, and RDKit/ChemNet versions: **XXX**;
- dataset, split, vocabulary, and canonicalization hashes: **XXX**;
- all hyperparameters, seeds, source draws, and coupling settings: **XXX**;
- compiled-program, Active8, sampler-stream, and recovery-checkpoint manifests: **XXX**;
- generated SMILES, per-event trajectories, and raw metric outputs: **XXX**;
- multi-seed summaries, confidence intervals, and failed-run disclosure: **XXX**.

K LARGE LANGUAGE MODEL USAGE STATEMENT

Large language models were used for drafting and code assistance. All mathematical claims, citations, implementation details, and empirical values require author verification. No generated reference or result is retained without checking a primary source or frozen experiment artifact.